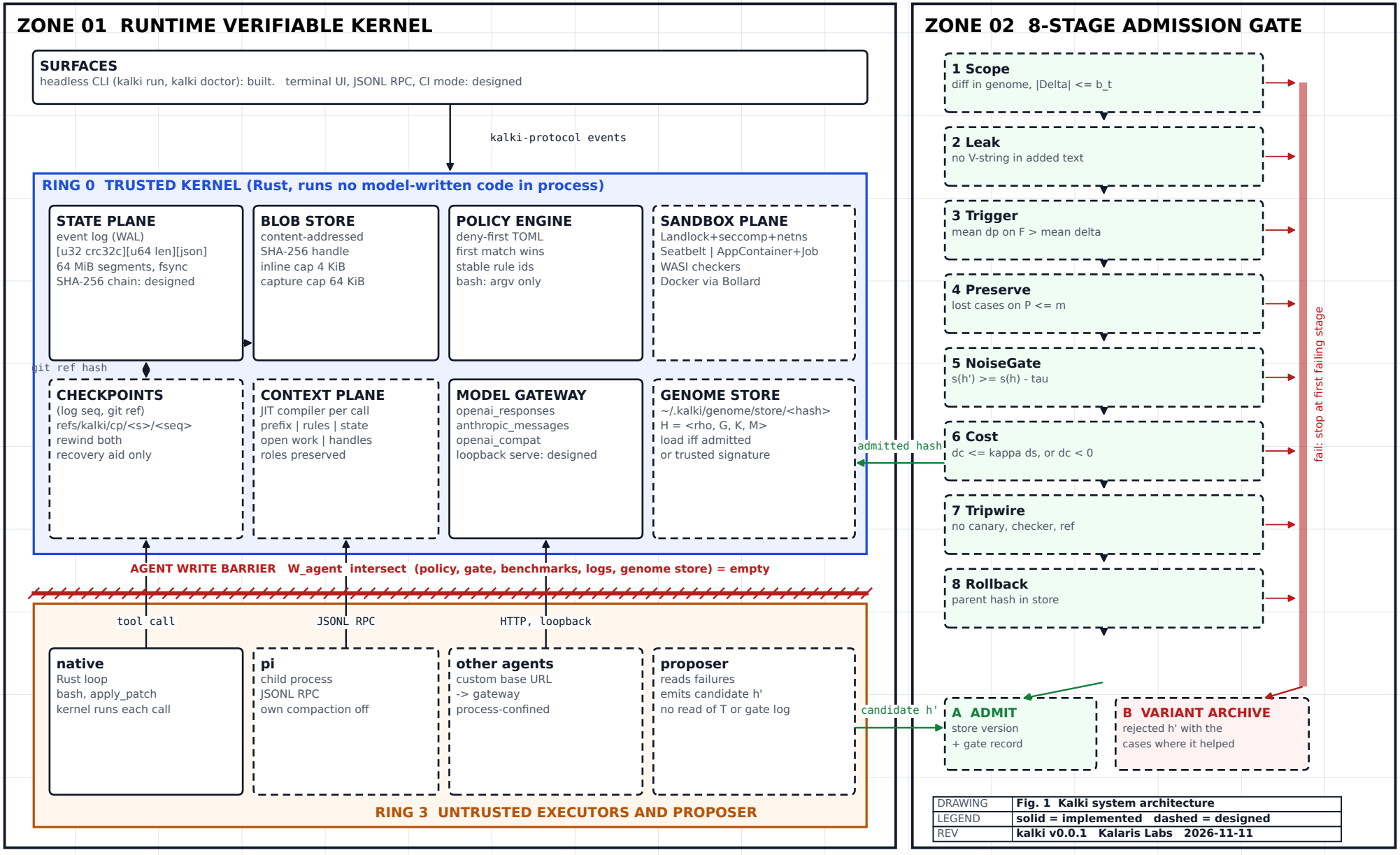
ZONE 01 RUNTIME VERIFIABLE KERNEL
SURFACES
headless CLI (kalki run, kalki doctor): built. terminal UI, JSONL RPC, CI mode: designed
kalki-protocol events
RING 0 TRUSTED KERNEL (Rust, runs no model-written code in process)
STATE PLANE
event log (WAL)
[u32 crc32c][u64 len][json]
64 MiB segments, fsync
SHA-256 chain: designed
BLOB STORE
content-addressed
SHA-256 handle
inline cap 4 KiB
capture cap 64 KiB
POLICY ENGINE
deny-first TOML
first match wins
stable rule ids
bash: argv only
SANDBOX PLANE
Landlock+seccomp+netns
Seatbelt | AppContainer+Job
WASI checkers
Docker via Bollard
git ref hash
CHECKPOINTS
(log seq, git ref)
refs/kalki/cp/<s>/<seq>
rewind both
recovery aid only
CONTEXT PLANE
JIT compiler per call
prefix | rules | state
open work | handles
roles preserved
MODEL GATEWAY
openai_responses
anthropic_messages
openai_compat
loopback serve: designed
GENOME STORE
~/.kalki/genome/store/<hash>
H = <rho, G, K, M>
load iff admitted
or trusted signature
admitted hash
AGENT WRITE BARRIER W_agent intersect (policy, gate, benchmarks, logs, genome store) = empty
tool call
JSONL RPC
HTTP, loopback
native
Rust loop
bash, apply_patch
kernel runs each call
pi
child process
JSONL RPC
own compaction off
other agents
custom base URL
-> gateway
process-confined
proposer
reads failures
emits candidate h'
no read of T or gate log
candidate h'
RING 3 UNTRUSTED EXECUTORS AND PROPOSER
ZONE 02 8-STAGE ADMISSION GATE
1 Scope
diff in genome, |Delta| <= b_t
2 Leak
no V-string in added text
3 Trigger
mean dp on F > mean delta
4 Preserve
lost cases on P <= m
5 NoiseGate
s(h') >= s(h) - tau
6 Cost
dc <= kappa ds, or dc < 0
7 Tripwire
no canary, checker, ref
8 Rollback
parent hash in store
fail: stop at first failing stage
A ADMIT
store version
+ gate record
B VARIANT ARCHIVE
rejected h' with the
cases where it helped
DRAWING
Fig. 1 Kalki system architecture
LEGEND
solid = implemented dashed = designed
REV
kalki v0.0.1 Kalaris Labs 2026-11-11